



FATE MONTHLY REPORT

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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
August 2024



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1 Introduction

The automatic Report has the goal to provide to ESO the performances assessment of the FATE project i.e. a picture of the characteristics of the forecasts delivered to ESO in operational configuration in the context of the FATE project. The report is supposed to be delivered to ESO with a monthly frequency. In agreement with ESO it has been decided to include in the report model performances following specific figures of merit as well as the success rate experienced by FATE, ESO or ECMWF during the process of forecast production and delivery. The three institutes represent the nodes of the automatic system. The figures of merit used to quantify the forecasts performances have been identified in agreement with ESO. Forecasts products delivered to ESO follow contractual specifications (SoW) and contract documentation.

For the night time FATE is supposed to provide forecasts related to 8 parameters: wind speed (WS at 30 m above the ground) and direction (WD at 30 m above the ground), relative humidity (RH at 30 m above the ground), precipitable water vapour (PWV), seeing (ε), wavefront coherence time (τ_0), ground layer fraction (GLF), sky transparency.

For the day time, FATE is supposed to provide forecasts related to 5 parameters: wind speed (WS at 30 m above the ground) and direction (WD at 30 m above the ground), relative humidity (RH at 30 m above the ground), precipitable water vapour (PWV), sky transparency.

The sky transparency forecast is provided with the operational General Circulation Model of ECMWF at high resolution (HRES). For all the other parameters FATE is supposed to provide forecasts at long time scale and at short time scale obtained with the Meso-Nh model and Astro-Meso-Nh model. The latter is the package for the optical turbulence i.e. C_N^2 and all integrated parameters depending on the C_N^2 . FATE allows different forecast time scales (FTS). Forecasts at short time scale are calculated using an autoregression approach (called AR hereafter) proposed in [1] and applied to the case of VLT [2]. The AR implies that, at each full time hour, when real-time observations start to be available FATE provides the forecasts for the coming 4h of all the three astroclimatic parameters (ε , τ_0 and GLF) plus the forecasts of four atmospheric parameters (WS, WD, RH and PWV). In other words, the forecasts are updated at each full time hour in the night and day time provided the conditions for the calculation are respected¹. We refer the reader to the paper [2] for the model configuration, the definition of parameters and the explanation of the AR method. Forecasts at long time scales are calculated with a time scale of 1 night, 2 nights, 3 nights as well as for 1 day, 2 days, 3 days and results are delivered 2 hours before sunset (for night time) and 2 hours before sunrise (for day time) following the configuration described in the document [4] located in the dedicated ESO repository². For the short time scales, in agreement with ESO, the report shows results related to the FTS at 1h that is the time scale more relevant for ESO for the VLT science operation.

Analysis of forecast performances is done using two different approaches:

- calculation of bias, RMSE and σ applied to the scatter plots between observations and forecasts. In [2] are reported definitions as well as the description of how data are treated.
- calculation of contingency tables (3×3 for all the parameters with exception of WD treated with a 4×4 table) and derived parameters: percentage of corrected forecasts (PC), extremely bad forecasts (EBD), probability of detection ($\text{POD}_{1,2,3}$). See [2]-Appendix and [5] for the definition and for an extended digression on these parameters.

Temporal frequency of sampling for observations and forecasts, in the standard configuration as well as in the AR case, is 10 minutes as requested by the ESO technical specifications.

¹We refer the reader to [3] for the these technicalities

²https://drive.google.com/drive/u/0/folders/1fa5r9EK8pfK1f18dFGxUmW_46OUyodIW. The repository is not public. Angel Otarola and Steffen Mieske are the ESO contact point for the FATE project and can access the repository.

LEGENDA

- **Standard configuration:** we mean with that those forecasts at long time scales (1 night/day, 2 nights/days, 3 nights/days) performed only with an atmospheric model having a forecast time scale (FTS) of [23h-33h], [47h-57h] and [71h-81h] respectively. Model configuration for the case 1 night is described in detail in [2] and [4]. Extrapolation to the other cases (2 nights/days and 3 nights/days) is reported in [4].

- **AR:** we mean with that the auto-regression method ([1, 2]). Such a method is a hybrid method taking into account the forecasts performed with the atmospheric model plus the real-time observations related to a finite number of nights in the close past. In [1] it has been proven that the optimal number of nights in the past is $N+5$. Both terms (observations and forecasts from the atmospheric model) are treated with a statistical algorithm, more precisely an auto-regression algorithm [1]. The method has been already applied in the forecast of the PWV at short time scale at VLT [6].

In the report data analysis is reported using the following logic. Each chapter of the report treats the cases: 1 night/day, 2 nights/days, 3 nights/days. In the case of 1 night/day the long FTS (standard configuration) as well as the short FTS (AR) are treated while, in the cases 2 nights/days and 3 nights/days, only the long FTS (standard configuration) is treated. Each main chapter has three sub-sections:

- The first sub-section (ex: 2.1) reports, for all the parameters, the scatter plots of the standard configuration, the AR at a time scale of 1h and the prediction by persistence a time scale of 1h. For the case of seeing, we considered observations only $< 1.5''$. This is because it is known that performances with the standard configuration decrease as soon as the value of the parameters increases. Due to the fact that AO can hardly be used for $\varepsilon > 1.5''$ it is considered useless (or weakly interesting) to analyse performances for seeing $> 1.5''$. Definition of the prediction by persistence is given in [7]. In synthesis, it assumes that the value of the parameter P at a time t (i.e. $P(t)$) for whatever $t > t_0$ is equal to the mean of P calculated in the 10 minutes before t_0 .
- In the second sub-section (ex: 2.2), values of bias, RMSE and σ are reported in tables of synthesis. To facilitate the analysis, tables show the statistics obtained in the last month as well as the statistics obtained since the beginning of the operational phase (this case is called 'incremental month'). The latter is obviously more representative from a statistical point of view and we expect that the latter case should approach the statistics obtained on a long time scale (12 months i.e. a whole solar year) in the context of a research approach [2]. The statistics performed on a time scale of the current month is less representative in statistical terms and it is possible to observe some fluctuations from one month to the other. This has been verified in the commissioning.
- The third sub-section (ex: 2.3) reports the following statistical operators values (PC, EBD, $POD_{1,2,3}$) retrieved from the contingency tables calculated for each atmospheric and astroclimatic parameter. In this section, as well as in the Annex, we take into consideration the 'incremental month' approach that means that we take into account data starting from the beginning of the operational phase up to the last month. For each parameters we used, as thresholds of the contingency tables, the first and second tertiles calculated off-line on a climatological time scale, more precisely on a time scale of 1 solar year [1/8/2018-31/7/2019] (the period used in the paper [2]). Values of first and second tertiles for all the parameters are reported in the documented [8]. The WD is the unique parameter that requires a 4×4 table. In that case the thresholds selected have been the directions: $[45^\circ-225^\circ]$ and $[135^\circ-315^\circ]$ identifying the sectors: North (N), East (E), South (S) and West (W) (see [5] Fig.3). In the day time case we introduced a correction on the thresholds [8] derived from an analysis carried out during the commissioning. In the calculation of the contingency tables, for the astroclimatic parameters case, (i.e. those parameters depending on the C_N^2), two different options are taken into account:

(A) the simple and classical contingency table in which are reported the number of couples (observation, forecast) falling in the each box. In that case the reference is given by observations provided by the an instrument or another one depending on the parameter. See [2] for details. In synthesis: the DIMM for the seeing, the MASS-DIMM for τ_0 and GLF.

(B) a modified contingency tables in which the coincidence in the main diagonal is done taking into account a realistic accuracy for the estimate of the individual parameter. The accuracy is estimated with the σ calculated between instruments measuring the same parameter at the same time. We refer the reader to [2] for an extended discussion on that topic. The uncertainties (or accuracies) have been calculated on a very rich statistical samples of observations. The FATE forecasts of astrophysical parameters are supposed to be used for the Service Mode and science operations of the whole VLT i.e. the four UTs as well as the VLTI therefore we are interested on the whole plateau of Cerro Paranal. That means that is indeed poorly representative an estimate that does not take into account an accuracy/uncertainty.

For the atmospheric parameters we used only case (A). In other words we assume an uncertainty that is approximated to zero. We have no elements at present that let us think that, also for the those parameters, an uncertainty different from zero might be relevant.

Contingency tables are reported in an Annex that is delivered to ESO too, jointly with the Report, on a monthly time scale.

NOTE

*At the end of the commissioning [2023/09/01 - 2023/12/31] we discovered an effect (that we call hereafter "border effect") that degrades performances of forecasts in AR configuration. The bad news is that performances of the AR mode appear therefore less good than what it has been estimated in [2] even if the substantial improvement with respect to the standard configuration is evident. The good news is that we identified during the commissioning the cause of this effect. The main priority of the R&D activity of the FATE project will be therefore focused on looking for a solution to this problem in order to improve these results. An explanation of this effect and an explanation of the causes of this effect is reported in [3]. To simplify the understanding we call "post-processing" the data analysis performed in post-processing as in [2], we call "real-time" the data analysis performed using the outputs *.csv delivered to ESO. In this report we show, at present, results obtained with the "post-processing" case. Differences between the two cases are indicated in [7].*

2 NIGHT1: Statistical analysis of forecast performances

2.1 NIGHT1: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

Figures in setion 2.1 are included in the range: 1-7.

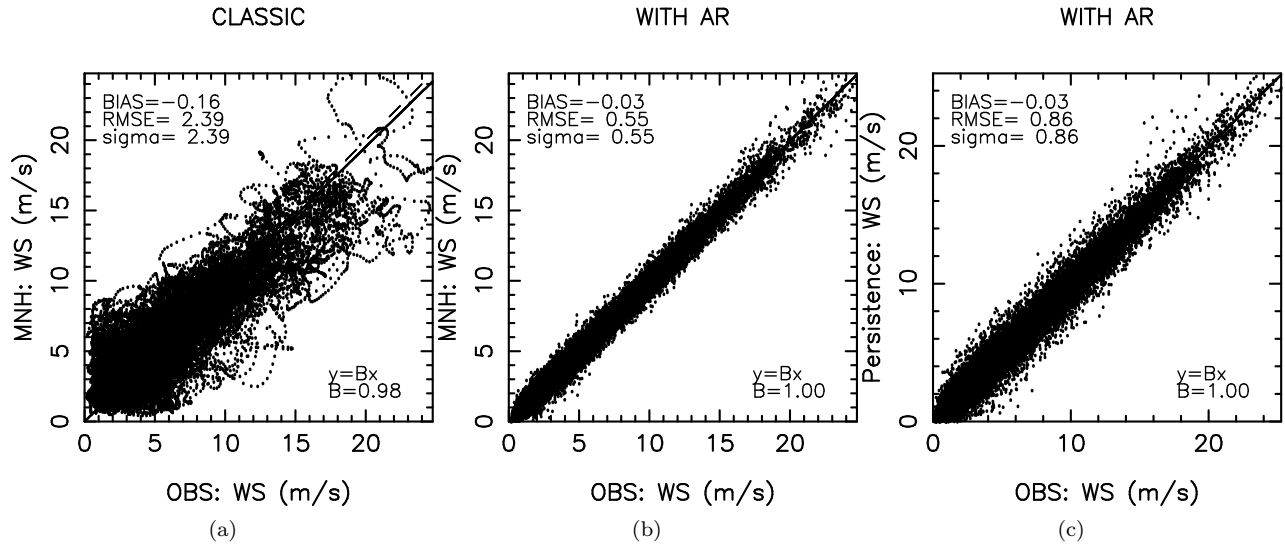


Figure 1: night1 - wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

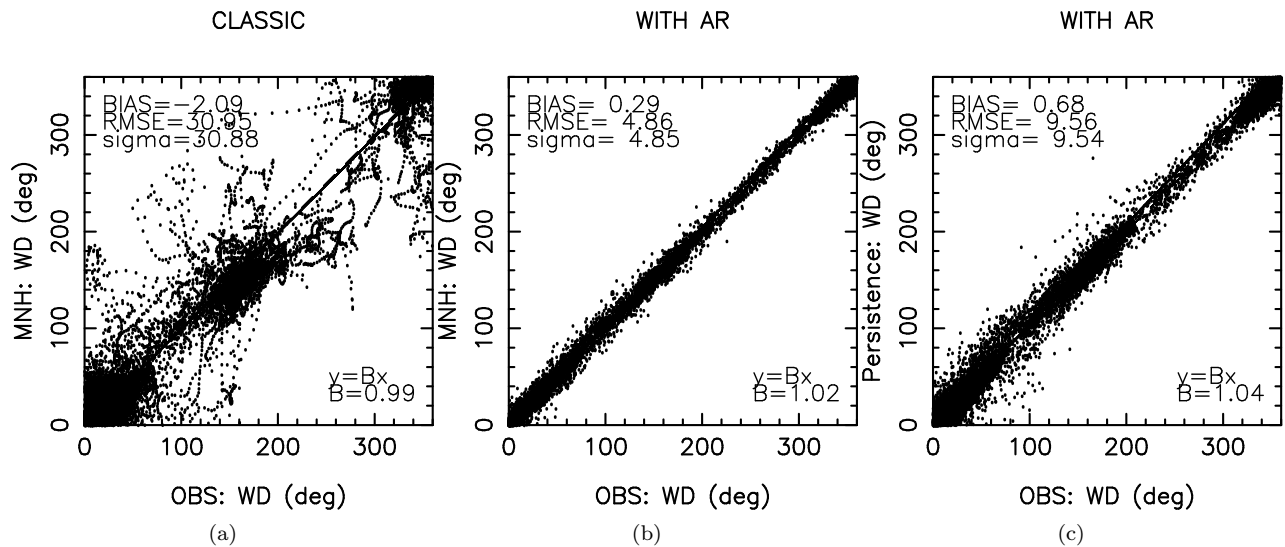


Figure 2: night1 - wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

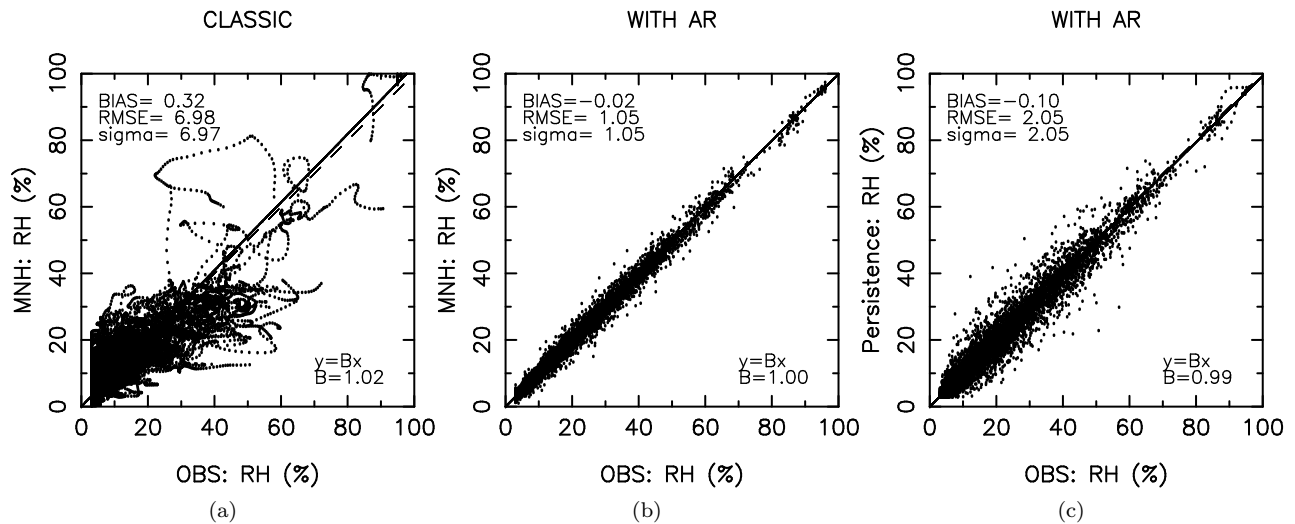


Figure 3: night1 - relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

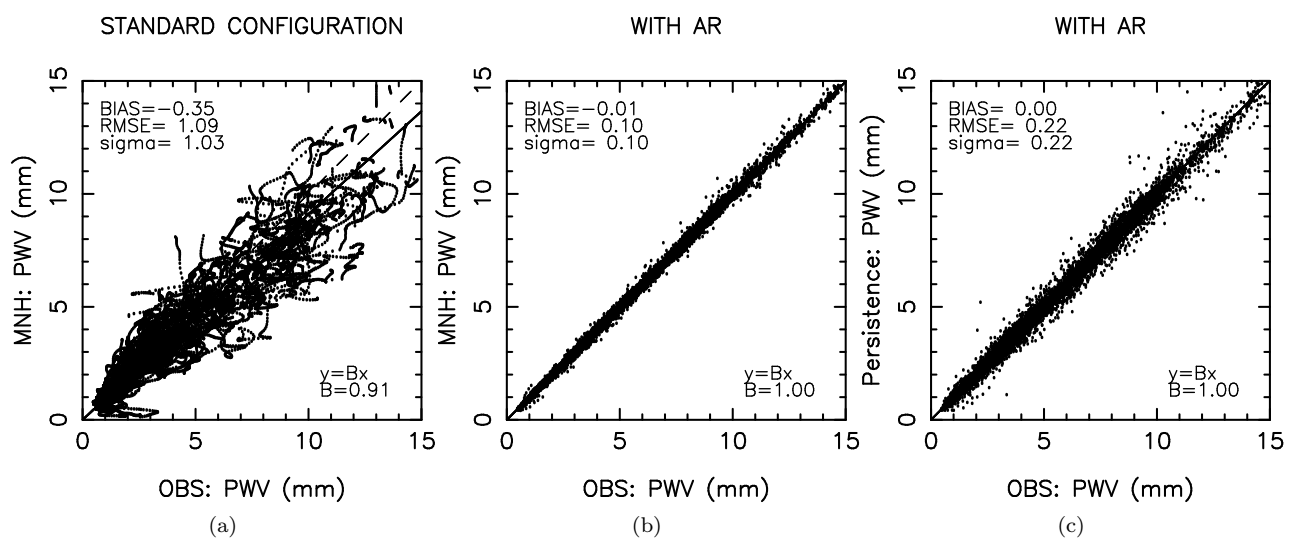


Figure 4: night1 - precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

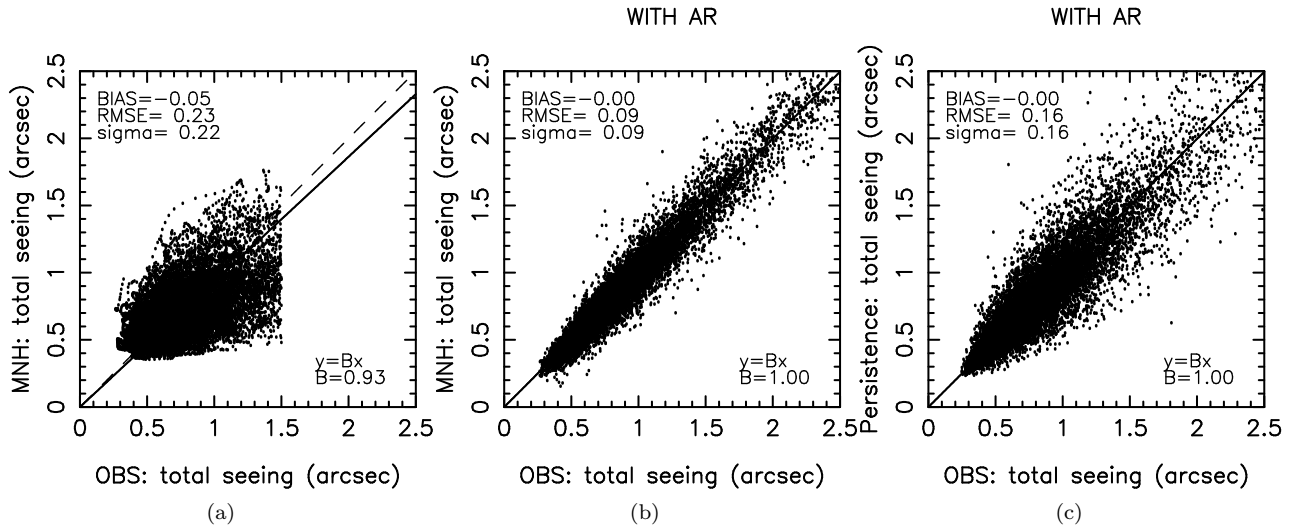


Figure 5: night1 - total seeing (arcsec): (a) STANDARD CONFIGURATION ($< 1.5''$), (b) WITH AR (1H), (c) PERSISTENCE (1H)

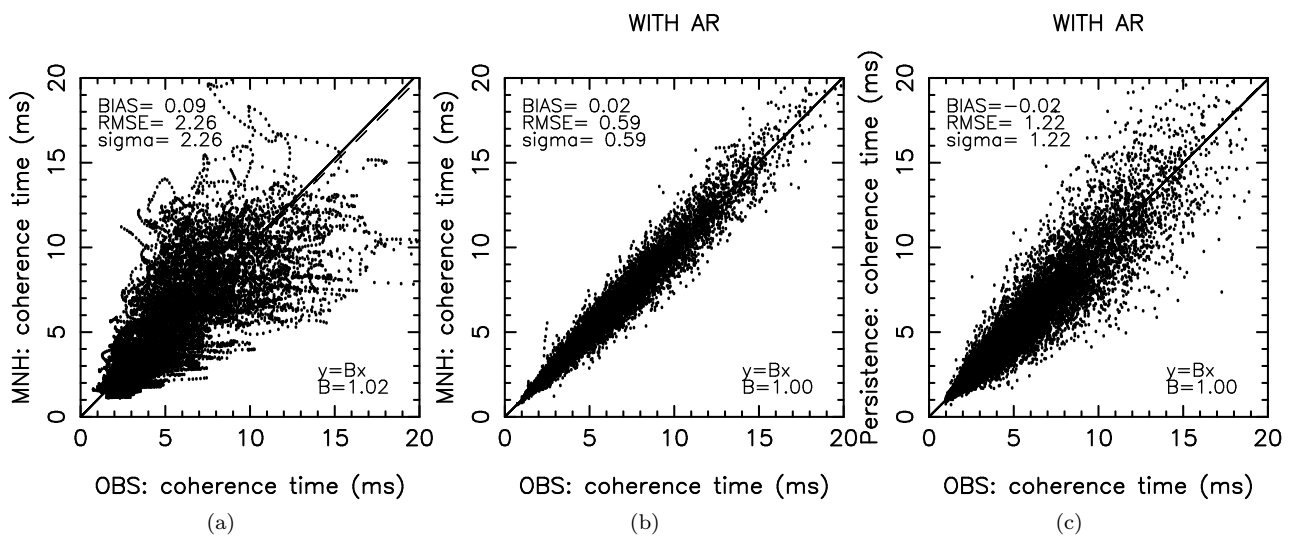


Figure 6: night1 - wavefront coherence time (ms): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

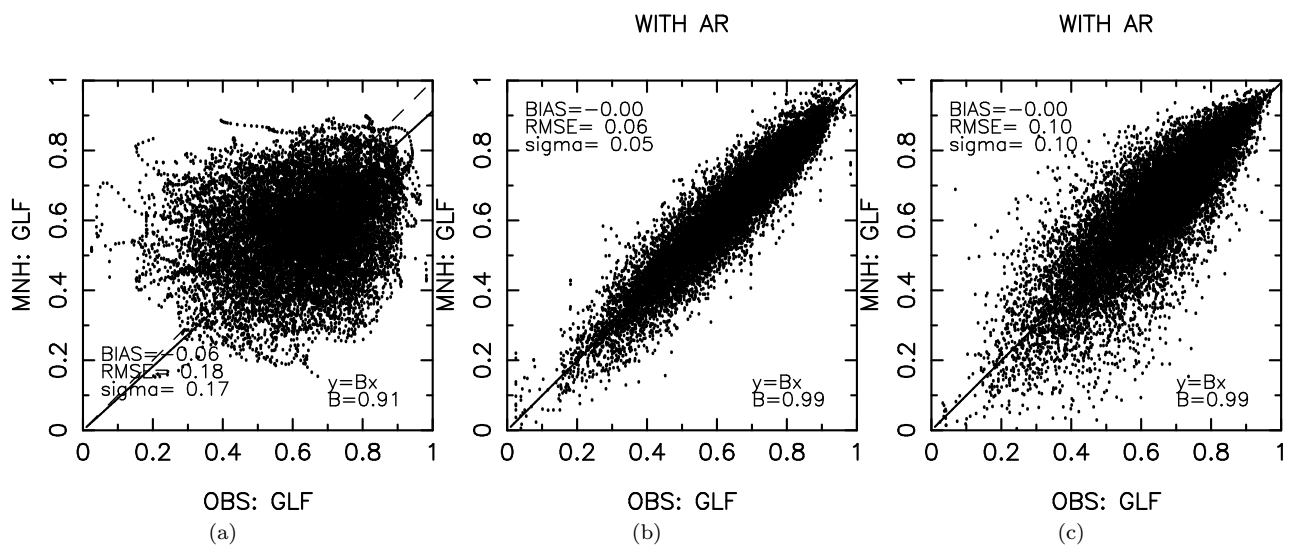


Figure 7: night1 - ground layer fraction (dimensionless): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

2.2 NIGHT1: Summarizing values of bias, RMSE, SD for long and short forecast time scales

Tables in section 2.2 are included in the range: 1-3.

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($m\cdot s^{-1}$)	-0.16	2.39	2.39	0.39	2.39	2.36
WD (degree)	-2.09	30.95	30.88	3.32	29.79	29.60
RH (percent)	0.32	6.98	6.97	0.45	9.52	9.51
PWV (mm)	-0.35	1.09	1.03	-0.39	0.77	0.67
SEE (arcsec)	-0.05	0.23	0.22	-0.08	0.23	0.21
TAU (ms)	0.09	2.26	2.26	-0.57	1.45	1.33
GLF (dimensionless)	-0.06	0.18	0.17	-0.06	0.16	0.15

Table 1: night1 - Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (August-2024)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m\cdot s^{-1}$)	0.01	0.57	0.57		0.91	0.91
WD (degree)	0.54	3.98	3.94		7.66	7.66
RH (percent)	-0.10	0.97	0.96		1.73	1.73
PWV (mm)	-0.01	0.07	0.07		0.20	0.20
SEE (arcsec)	0.00	0.09	0.09		0.16	0.16
TAU (ms)	-0.02	0.30	0.30		0.56	0.56
GLF (dimensionless)	-0.00	0.05	0.05		0.09	0.09

Table 2: night1 - Last month (August-2024): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m\cdot s^{-1}$)	-0.03	0.55	0.55		0.86	0.86
WD (degree)	0.29	4.86	4.85		9.56	9.54
RH (percent)	-0.02	1.05	1.05		2.05	2.05
PWV (mm)	-0.01	0.10	0.10		0.22	0.22
SEE (arcsec)	-0.00	0.09	0.09		0.16	0.16
TAU (ms)	0.02	0.59	0.59		1.22	1.22
GLF (dimensionless)	-0.00	0.06	0.05		0.10	0.10

Table 3: night1 - Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

2.3 NIGHT1: Probability of detection (PODi) for long and short forecast time scales

Tables in section 2.3 are included in the range: 4-15.

PARAMETER	STANDARD	WITH AR (1H)
POD1	64.5	93.9
POD2	50.2	88.5
POD3	82.5	96.2
PC	67.5	93.3
EBD	1.7	0.0

Table 4: night1 - PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	98.7	99.3
POD2	54.9	89.3
POD3	15.5	83.4
PC	91.4	97.7
EBD	0.1	0.0

Table 5: night1 - PODs for wind speed (ms^{-1}) and thresholds (12.0,18.0) ms^{-1}

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PARAMETER	STANDARD	WITH AR (1H)
POD1	91.3	98.8
POD2	53.7	91.2
POD3	81.9	98.7
POD4	35.9	93.4
PC	83.5	97.8
EBD	1.2	0.0

Table 6: night1 - PODs for wind direction (degree)

PARAMETER	STANDARD	WITH AR (1H)
POD1	52.1	93.4
POD2	60.9	93.8
POD3	83.1	97.2
PC	64.4	94.7
EBD	2.7	0.0

Table 7: night1 - PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	99.5	99.9
POD2	33.6	86.6
POD3	77.0	96.1
PC	98.3	99.7
EBD	0.1	0.0

Table 8: night1 - PODs for relative humidity (percent) and thresholds (50,70) percent

PARAMETER	STANDARD	WITH AR (1H)
POD1	88.8	98.3
POD2	NaN	96.7
POD3	92.6	99.0
PC	91.4	98.1
EBD	8.6	0.0

Table 9: night1 - PODs for precipitable water vapor (mm)

PARAMETER	STANDARD	WITH AR (1H)
POD1	60.2	88.7
POD2	41.0	76.6
POD3	48.0	90.7
PC	49.3	85.5
EBD	6.9	0.0

Table 10: night1 - PODs for total seeing (arcsec) and accuray 0.0

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PARAMETER	STANDARD	WITH AR (1H)
POD1	95.0	100.0
POD2	97.0	99.8
POD3	83.8	99.9
PC	94.3	99.9
EBD	4.7	0.0

Table 11: night1 - PODs for total seeing (arcsec) and accuray 0.24

PARAMETER	STANDARD	WITH AR (1H)
POD1	76.8	93.9
POD2	42.5	86.7
POD3	83.8	94.5
PC	68.2	91.8
EBD	1.5	0.0

Table 12: night1 - PODs for wavefront coherence time (ms) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	92.9	99.9
POD2	76.3	99.8
POD3	84.2	99.8
PC	87.0	99.9
EBD	3.9	0.0

Table 13: night1 - PODs for wavefront coherence time (ms) and accuray 1.22

PARAMETER	STANDARD	WITH AR (1H)
POD1	53.4	87.2
POD2	43.1	73.1
POD3	23.2	85.2
PC	39.5	81.5
EBD	14.3	0.1

Table 14: night1 - PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	88.1	99.8
POD2	88.0	99.8
POD3	49.7	99.7
PC	80.0	99.8
EBD	15.2	0.1

Table 15: night1 - PODs for ground layer fraction (dimensionless) and accuray 0.14

3 DAY1: Statistical analysis of forecast performances related

3.1 DAY1: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

Figures in section 3.1 are included in the range: 8-11.

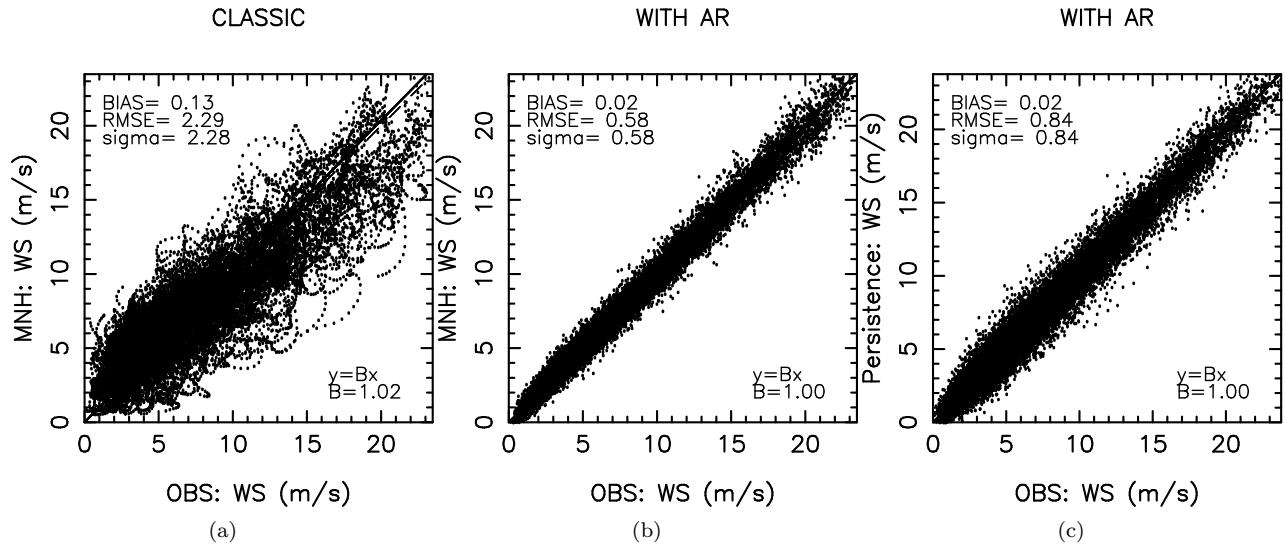


Figure 8: day1 - wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

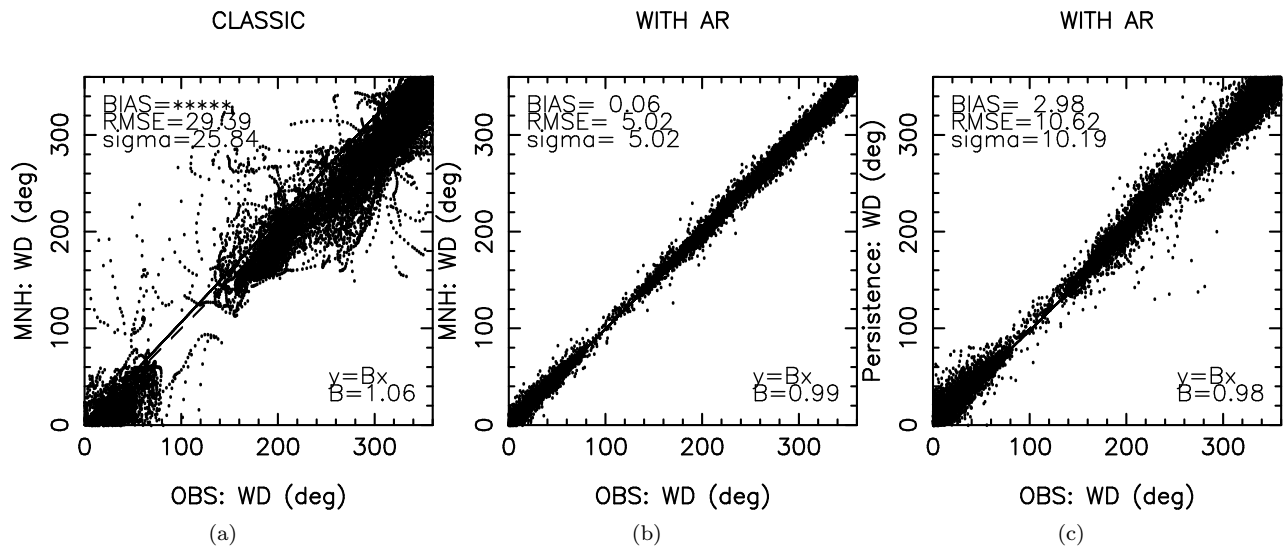


Figure 9: day1 - wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

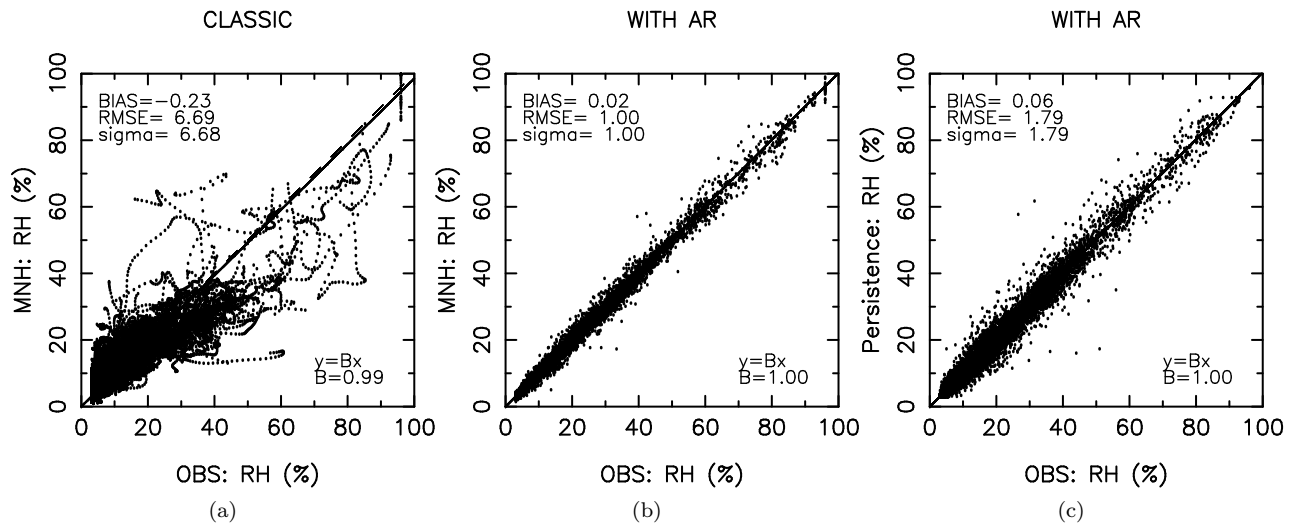


Figure 10: day1 - relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

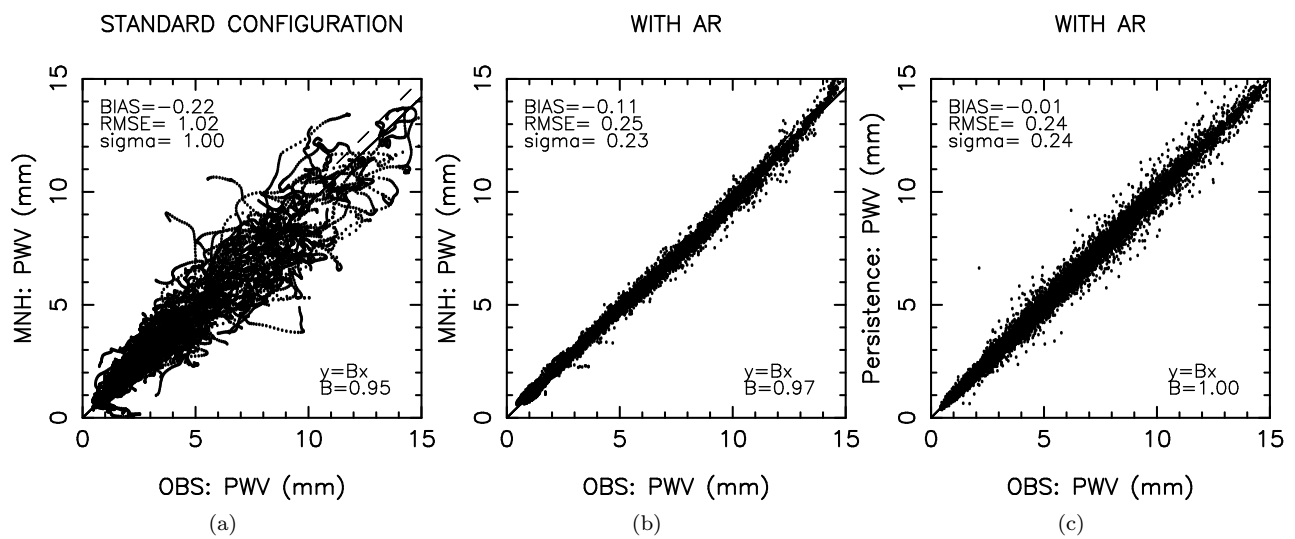


Figure 11: day1 - precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

3.2 DAY1: Summarizing values of bias, RMSE, SD for long and short forecast time scales

Tables in section 3.2 are included in the range: 16-18.

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($m \cdot s^{-1}$)	0.13	2.29	2.28	-0.60	2.50	2.43
WD (degree)	-14.00	29.39	25.84	-10.94	25.27	22.78
RH (percent)	-0.23	6.69	6.68	0.90	6.15	6.08
PWV (mm)	-0.22	1.02	1.00	-0.21	0.76	0.73

Table 16: day1 - Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (August-2024)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.03	0.64	0.64		0.85	0.85
WD (degree)	0.42	4.47	4.45		8.79	8.50
RH (percent)	0.09	0.87	0.87		1.48	1.48
PWV (mm)	0.05	0.15	0.14		0.18	0.18

Table 17: day1 - Last month (August-2024): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	0.02	0.58	0.58		0.84	0.84
WD (degree)	0.06	5.02	5.02		10.62	10.19
RH (percent)	0.02	1.00	1.00		1.79	1.79
PWV (mm)	-0.11	0.25	0.23		0.24	0.24

Table 18: day1 - Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

3.3 DAY1: Probability of detection (PODi) for long and short forecast time scales

Tables in section 3.3 are included in the range: 19-24.

PARAMETER	STANDARD	WITH AR (1H)
POD1	50.2	91.3
POD2	55.4	88.1
POD3	82.3	95.5
PC	64.4	92.0
EBD	1.7	0.0

Table 19: day1 - PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	98.0	99.0
POD2	52.1	89.1
POD3	36.8	87.5
PC	89.5	97.3
EBD	0.2	0.0

Table 20: day1 - PODs for wind speed (ms^{-1}) and thresholds (12.0,18.0) ms^{-1}

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PARAMETER	STANDARD	WITH AR (1H)
POD1	74.8	96.7
POD2	35.6	92.3
POD3	89.9	97.4
POD4	80.8	94.9
PC	76.6	96.1
EBD	0.5	0.0

Table 21: day1 - PODs for wind direction (degree)

PARAMETER	STANDARD	WITH AR (1H)
POD1	55.8	95.0
POD2	67.8	94.5
POD3	81.1	97.3
PC	67.5	95.5
EBD	1.0	0.0

Table 22: day1 - PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	99.7	99.9
POD2	29.1	94.8
POD3	38.7	92.0
PC	97.9	99.7
EBD	0.2	0.0

Table 23: day1 - PODs for relative humidity (percent) and thresholds (50,70) percent

PARAMETER	STANDARD	WITH AR (1H)
POD1	82.4	96.8
POD2	NaN	95.8
POD3	94.9	98.3
PC	91.0	97.1
EBD	9.0	0.0

Table 24: day1 - PODs for precipitable water vapor (mm)

References

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